Question 5

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make function for displaying menu
 Function display ()
  Function register member ( )
    Take input for name , Student ID
      and member type
    Validate the input member type.
    Print "registerred successfully" if all the
    inputs got valid
END Function
Function display books ( ~~&~~ book ID~~&~~ [ ], book title [ ][50],
                                available [ ], total )
    ~~Print ( )~~ for(~~i=~~int i=0; i< total; i++)
            print ( book ID [i], book title[i][50]
                available [i] )//printing all the
                book ids, titles and available
                copies
Function borrow ( book ID[], book title [ ][50], available [ ],
                total, borrowed ID [ ], ~~&~~ Count, type )
 • Int id, days, idx = -1, limit
  • ~~int lim~~ if member type is 1 then limit = 14 else
     limit = 30 //END IF
  • ~~&~~ Borrow count should be less than 3 to
  • Take input for book id
  • Check for the id in array of book ids and
    save it's index in idx and print invalid
    id if not found.
    • ~~Availab~~ Check that book bra should be available
      if not end function with a message.
    • Take input number of days book will be borrowed.
    • Validate the input that days should be positive.
    • Borrowed ID [ count + 1 ] = id
    • available [ idx ] = available [ idx ] - 1
    • Check that the days should not exceed the limit
END Function
Function return Book( Book ID[ ], available [ ], total,
                     borrowed ID[ ], count, type, &Tfine )
    , int id, day, & pos = -1, idx = -1, extra, limit, rate
    • char code [20], correct [ ] = "LIB2026"
    • float fine = 0
    • Take input for id
    • Search for the id in borrowed ID [ ] and if not
      found then end function.
    • Take input for days and validate it.
    • Search for the id in Book ID and store it's index in
      idx and increment the value in available [ ]
    • if type == 1 then set limit to 14 else set limit to 30
    • if type == 1 then set rate to 20 else set rate to 10
    • if ( days > ~~the~~ limit ) {
          extra = day - limit
          fine = extra * rate
          Tfine = Tfine + fine
          Take input for membership code
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        for (int i = 0 ; correct[i] != '\0' AND code[i] == correct[i] ; i++)
        END FOR
        if correct[i] == '\0' AND code[i] == '\0' {
            fine = fine / 2 } END IF
        ~~else~~
        print (fine) // printing fine for this return
        } ~~ENDIF~~
       ~~e~~ELSE print("Book returned on time")
END Function
Function display Record( name[], SID, type, borrowed ID[],
                          count, TFine, D fine )
   print( name, SID, type, count, Tfine, D fine )
   // printing name and ID and is the person is student
   // or faculty, his borrowed books, their fine and fine
   // after discount *
 END Function
  main
  Initialize book ID array
  Initialize book title array
   Initialize available array
   Initialize total
   char name [50] = " "
    int SID = 0 type = 0 , registered = 0
    int Borrowed id [3] , count = 0
    float Tfine = 0 , D fine = 0
```

Date:

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do {
    display( )
    Take input the choice
    Switch (choice) {
        1: Check for registration if not registered
           register him through function
        2: Display book (book ID, book title, available,
           issue )
        3: Call borrow Book function
        4: Call return book function and add it in D.
           fun
        5: ~~Displ~~. Call Display record function
        6: Exit the program
} while (choice is ~~not~~ != 6)
```

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